

Submission deadline: **April 5, 12:00 AM**

Task 1: Data Preparation

1. Load the dataset from the class GitHub page
2. Preprocess the data
 - Drop any missing values.
 - Convert categorical features into the numerical form using one-hot encoding.
 - Standardize the numerical features using StandardScaler.
 - Split the data into training, validation, and test sets.

Task 2: Building and Training MLP with Keras Sequential API

1. Build a basic MLP using the Keras sequential model with different numbers of hidden layers and units (at least two different models).
2. Use the 'ReLU' Activation function for hidden layers and 'Softmax' for the output layer.
3. Set a proper metric to compile your model with.
4. Train the model with Adam optimizer and set the learning rate.
5. Plot loss and metric during training.
6. Evaluate the models' accuracy, compare and analyze results.
7. Choose the best architecture for the following tasks.

Task 3: Building and Training MLP with Keras Functional API

1. Define a multi-input architecture as Figure 1 where:
 - Categorical features go through a stack of N Dense layers.
 - Numerical features go through a separate stack of M Dense layers.
 - These two feature paths are concatenated before passing through the final K Dense layers.
2. Use 'ReLU' Activation function for hidden layers
3. Set a proper activation function for the output layer of categorical and numerical features networks and explain your choice.
4. Design your network with an arbitrary number of N, M, and K.
5. Train the model with Adam optimizer and evaluate its accuracy.
6. Compare the results with sequential networks in Task 2.

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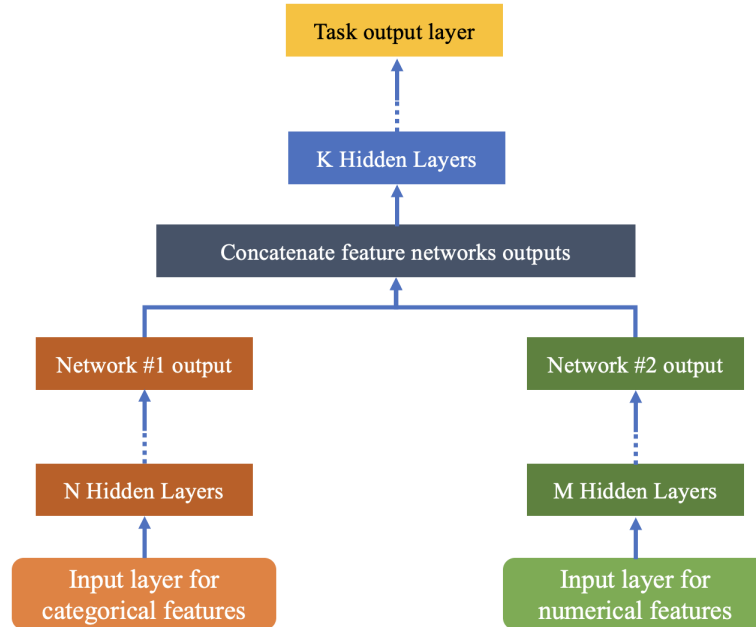


Figure 1: Task 3 Architecture

Task 4: Exploring Optimization Algorithms

Train the Sequential model with the following optimizers and compare the results in the context of speed, accuracy, etc.:

- SGD (with and without momentum)
- Nadam
- Adagrad
- RMSProp

Task 5: Preventing Overfitting

Train the Functional API model with these techniques and compare results:

- L1 and L2 Regularization
- Dropout (0.3 and 0.1 probability in hidden layers)

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Task 6: Exploring Different Activation Functions

Choose 2 of the following activation functions and train the sequential model using different activation functions in the hidden layers and compare your results with Task2:

- Sigmoid
- Tanh
- Leaky ReLU
- ELU
- Swish

Extra Point: Using TensorBoard to compare the different models trained in the previous tasks has extra points.

Submission Guidelines

- Homeworks must be completed in **groups of four**.
- Each group must submit a single file containing a **PDF report**, including answers to discussion questions, visualizations, and code explanations, and a **Jupyter Notebook**, containing all code and results.